Signed on as **ฉัตรพล โอวานนท์ชัย**

My worksheet

Quiz 1: Database and SQL

Out: Sep 2 2026 — Due: Sep 11 2026 23:59

2107540 (2026/1)
For individual student

Submissions CANNOT be made after the due date/time.(11-Sep-2026 23:59)

Submission can be made ONLY ONCE.

Instruction

undefined

My Work



Complete the question set

1 point

1 Which statement best describes the purpose of a Primary Key?

Pick a choice:

- ◯ **To store numerical values**
- ◯ **To uniquely identify each record in a table**
- ◯ **To connect every table automatically**
- ◯ **To sort records in a table**

1 point

2 In the following tables, which attribute should be a Foreign Key in `company_perf` ?

company_info
company_id (PK)
company_name
industry

company_perf

performance_id
company_id
year
esg_overall
carbon_emissions

Pick a choice:

- **performance_id**
- **company_id**
- **year**
- **esg_overall**

1 point

3

Each monitoring record belongs to exactly one station. What is the cardinality?

Pick a choice:

- **One-to-One**
- **One-to-Many**
- **Many-to-Many**
- **Many-to-One**

1 point

4

Consider the following table:

COMPANY_PERFORMANCE
Performance_ID (PK)
Company_ID
Company_Name
Industry
Year
ESG_Overall

A company can have many performance records across different years. Which change would **best improve the database design**?

Pick a choice:

- **Move Year to the COMPANY table**
- **Remove Performance_ID and use Company_ID as the Primary Key**
- **Duplicate Company_Name and Industry for every performance record**
- **Move Company_Name and Industry to a separate COMPANY table**

1 point

What is the main purpose of normalization?

5

Pick a choice:

- ◯ **To make tables larger**
- ◯ **To increase the number of duplicate values**
- ◯ **To reduce data redundancy and prevent update anomalies**
- ◯ **To make SQL queries unnecessary**

1 point

6

Which SQL statement correctly retrieves companies from the `Asia` region?

Pick a choice:

- ◯ **SELECT * FROM company_info WHERE region = 'Asia';**
- ◯ **SELECT * FROM company_info IF region = 'Asia';**
- ◯ **GET * FROM company_info WHERE region = 'Asia';**
- ◯ **SELECT company_info WHERE region = 'Asia';**

1 point

7

Which SQL statement returns the **10 companies with the highest carbon emissions**?

Pick a choice:

- ◯ **SELECT * FROM company_perf ORDER BY carbon_emissions LIMIT 10;**
- ◯ **SELECT * FROM company_perf ORDER BY carbon_emissions DESC LIMIT 10;**
- ◯ **SELECT TOP 10 * FROM company_perf SORT BY carbon_emissions;**
- ◯ **SELECT * FROM company_perf GROUP BY carbon_emissions LIMIT 10;**

1 point

8

What does `GROUP BY year` do in the following query?

```
SELECT year,
        AVG(esg_environmental)
FROM company_perf
GROUP BY year;
```

Pick a choice:

- ◯ **Sorts the data by year**
- ◯ **Removes the year column**
- ◯ **Groups records by year before calculating the average**
- ◯ **Calculates the average esg_environmental score**

1 point

9

What is the purpose of the `JOIN` in this query?

```
SELECT c.company_name,
       p.esg_environmental
FROM company_info AS c
JOIN company_perf AS p
    ON c.company_id = p.company_id;
```

Pick a choice:

- ◯ **To delete duplicate companies**
- ◯ **To sort companies by esg_environmental score**
- ◯ **To combine related information from two tables**
- ◯ **To calculate the average esg_environmental score**

1 point

10

Which SQL query correctly finds industries with an **average ESG Environmental score greater than 65**?

Pick a choice:

- ◯ **SELECT industry, AVG(esg_environmental) FROM company_perf WHERE AVG(esg_environmental) > 65 GROUP BY industry;**
- ◯ **SELECT industry, AVG(esg_environmental) FROM company_perf GROUP BY industry HAVING AVG(esg_environmental) > 65;**
- ◯ **SELECT industry, AVG(esg_environmental) FROM company_perf GROUP BY industry WHERE esg_environmental > 65;**
- ◯ **SELECT industry, AVG(esg_environmental) FROM company_perf HAVING esg_environmental > 65;**

(Optional) Call for the attention of

Show staff searching box

Submit your work (Submission can be made ONLY ONCE)

Save / Submit

Staff's Feedback

-- No feedback to this submission has been made by any of the course staffs yet. --